Part 1.

$A=1$, $A'=0$

$$AB'CD' + AB'C'D + ABCD + AC'D + A'BC'D$$

$$AB'CD' + AB'C'D + AB'C'D' + ABCD + ABC'D + AB'C'D + A'BC'D$$

$AB'CD'$	$AB'C'D$	$AB'C'D'$	$ABCD$	$ABC'D$	$AB'C'D$	$A'BC'D$
1010	1001	1000	1111	1101	1001	0101
10	9	8	15	13	9	5

$\sum m(5,8,9,10,13,15)$ total minterms = 6

Part 2

Simplify expression SOP

3)

$$AB'CD + AB'C' + ABCD + AC'D + A'BC'D$$

$$AB'(CD' + C') \rightarrow AB'(C' + D')$$

$$= AB'(C' + D') = AB'C' + AB'D'$$

$$AB'C' + AB'D' + ABCD + AC'D + A'BC'D$$

$$AB'C' = AB'C'(D + D') = \cancel{AB'C'D} + \cancel{AB'C'D'}$$

consensus Laws

Laws:

$$AB'D' = \cancel{AB'C'D'} + AB'CD'$$

$$AC'D = \cancel{AB'C'D} + ABC'D$$

$$AB'D' + \underline{ABCD + AC'D} + A'BC'D$$ absorption Law

$$AD(BC + C') + (C + C')(B + C') \rightarrow 1\cdot(B + C')$$

$$AD(B + C')$$

$$AB'D' + ABD + AC'D + A'BC'D$$

$AB'D'$	ABD	$AC'D$	$A'BC'D$
100	111	101	0101

4)

$$AB'CD' + AB'C' + ABCD + AC'D + A'BC'D$$

CD \ AB	00	01	11	10
00	0	4	12	1 (8)
01	1	1 (5)	1 (13)	1 (9)
11	3	7	1 (15)	11
10	4	6	14	1 (10)

$AB'C'D' + AB'C'D \rightarrow AB'C'$

$AB'C'D' + AB'CD' \rightarrow AB'D'$

$ABC'D + ABCD \rightarrow ABD$

$A'BC'D + ABC'D \rightarrow BC'D$

$$F_s = AB'C' + AB'D' + ABD + BC'D$$

Truth Table

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1 5
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1 8
1	0	0	1	1 9
1	0	1	0	1 10
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1 13
1	1	1	0	0
1	1	1	1	1 15

A = 0 – 800ns = 0
A = 900 – 1500ns = 1

B = 0 = 0
4

$A=1$, $A'=0$
$B=1$ $B'=0$
$C=1$ $C'=0$
$D=1$ $D'=0$

$$AB'C' + AB'D' + ABD + BC'D$$

Minterm Truth Table

	A	B	C	D	F
m5	0	1	0	1	1
m8	1	0	0	0	1
m9	1	0	0	1	1
m10	1	0	1	0	1
m13	1	1	0	1	1
m15	1	1	1	1	1